

PurpleAir
live sensors

Open-Meteo
weather

NOAA HMS
smoke

CAMS
AOD, PM2.5

EJScreen,
census tracts

38-feature pipeline
(BallTree neighbor PM2.5 at 25/50/100 km)

RF + LightGBM + CatBoost simplex ensemble
(**LOSO $R^2 = 0.714$, 310 sensors**)

live tract map
6,896 TX census tracts

exact SHAP blend
7 concept groups (this paper)